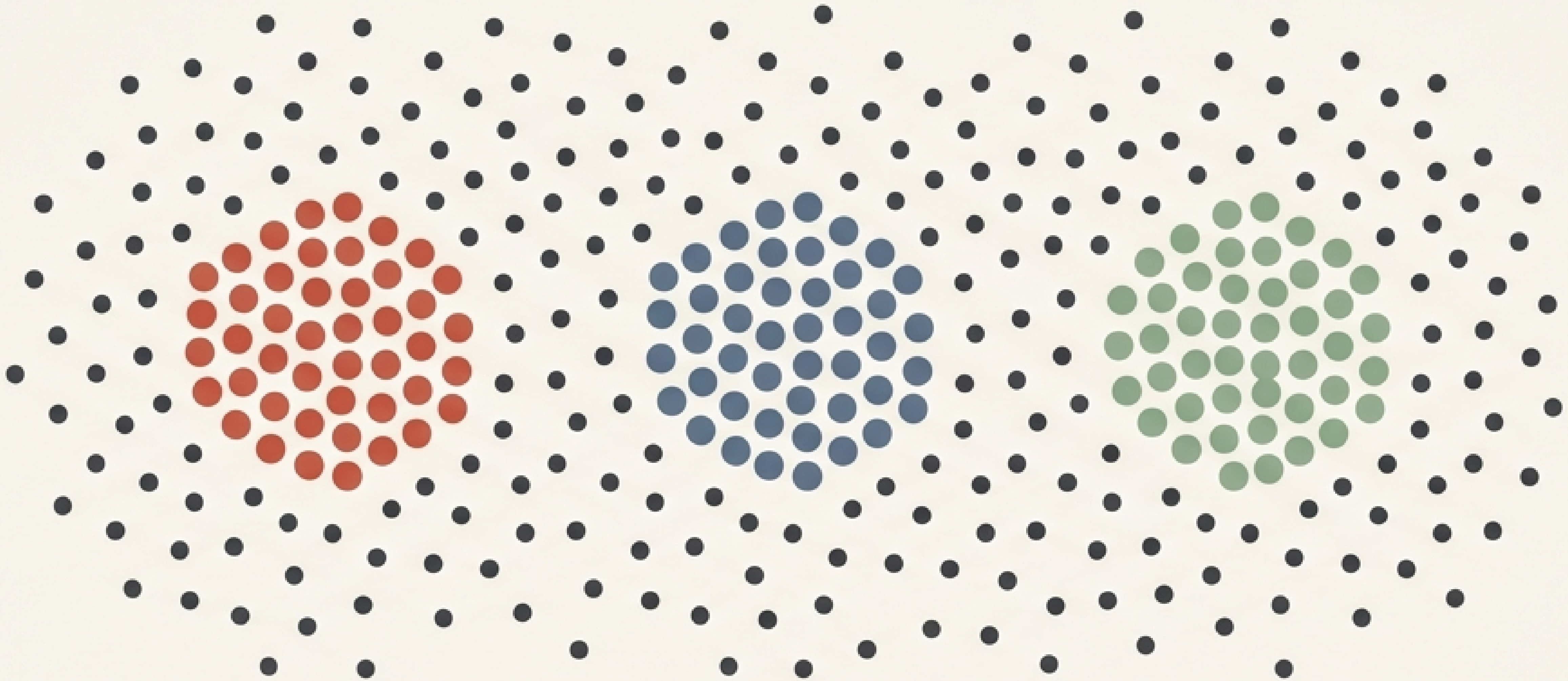


# Demystifying K-Means Clustering

Geometric Intuition for Unsupervised Learning

# Finding structure in unlabelled data

Customer Segmentation

Image Compression

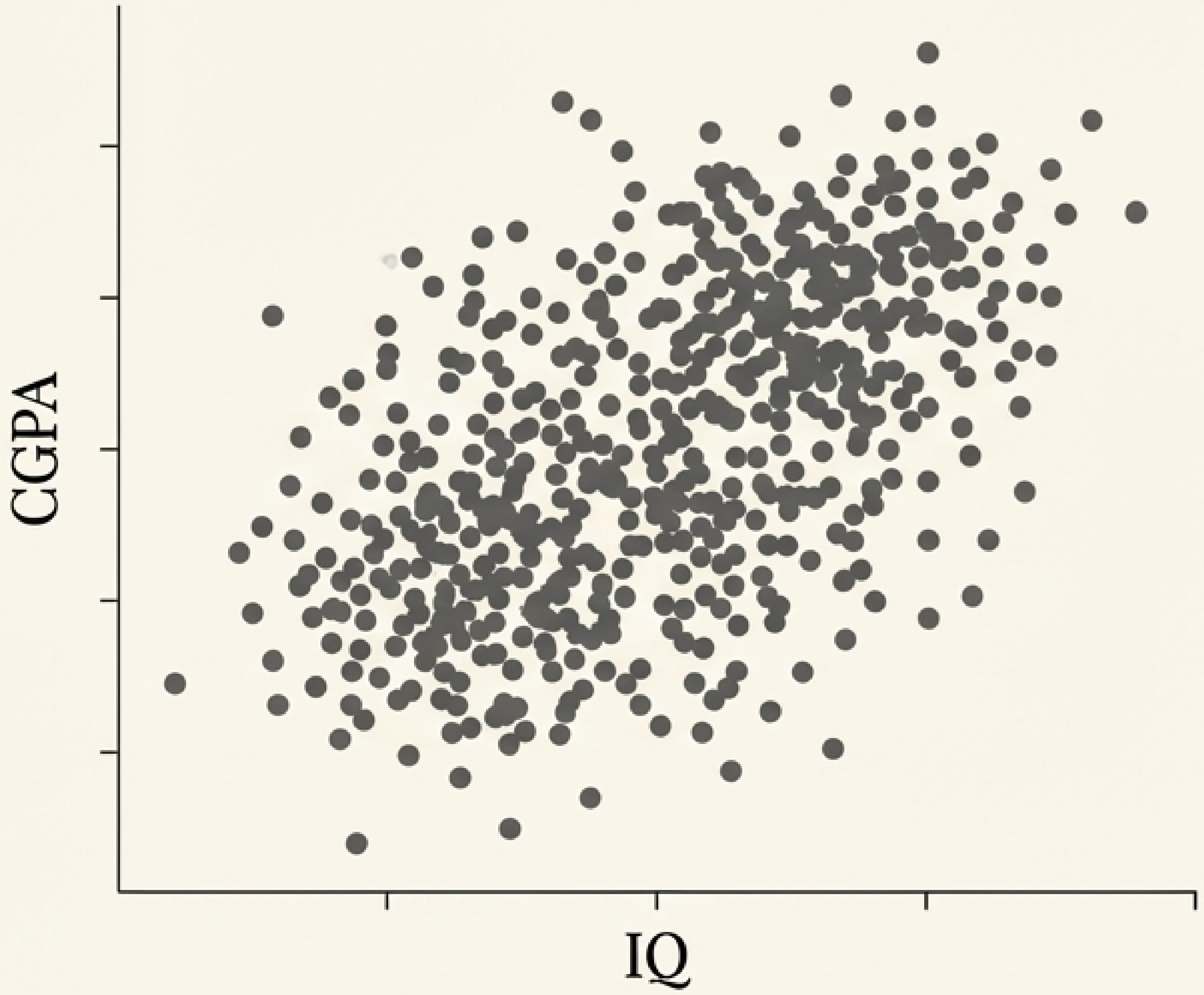
Student Profiling

Unsupervised Learning: Grouping similar data points together without pre-existing labels to uncover hidden patterns.

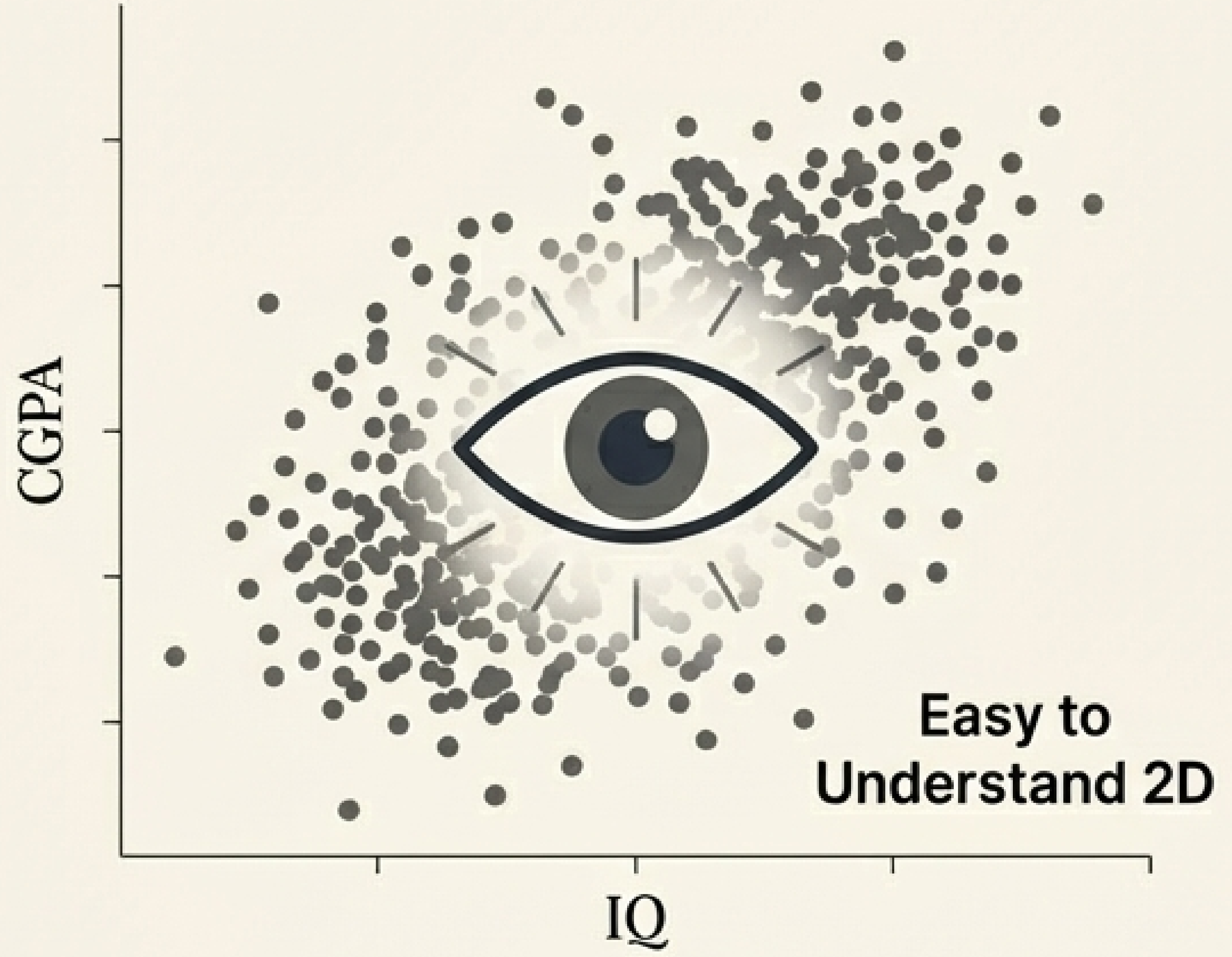
# The college placement cell challenge

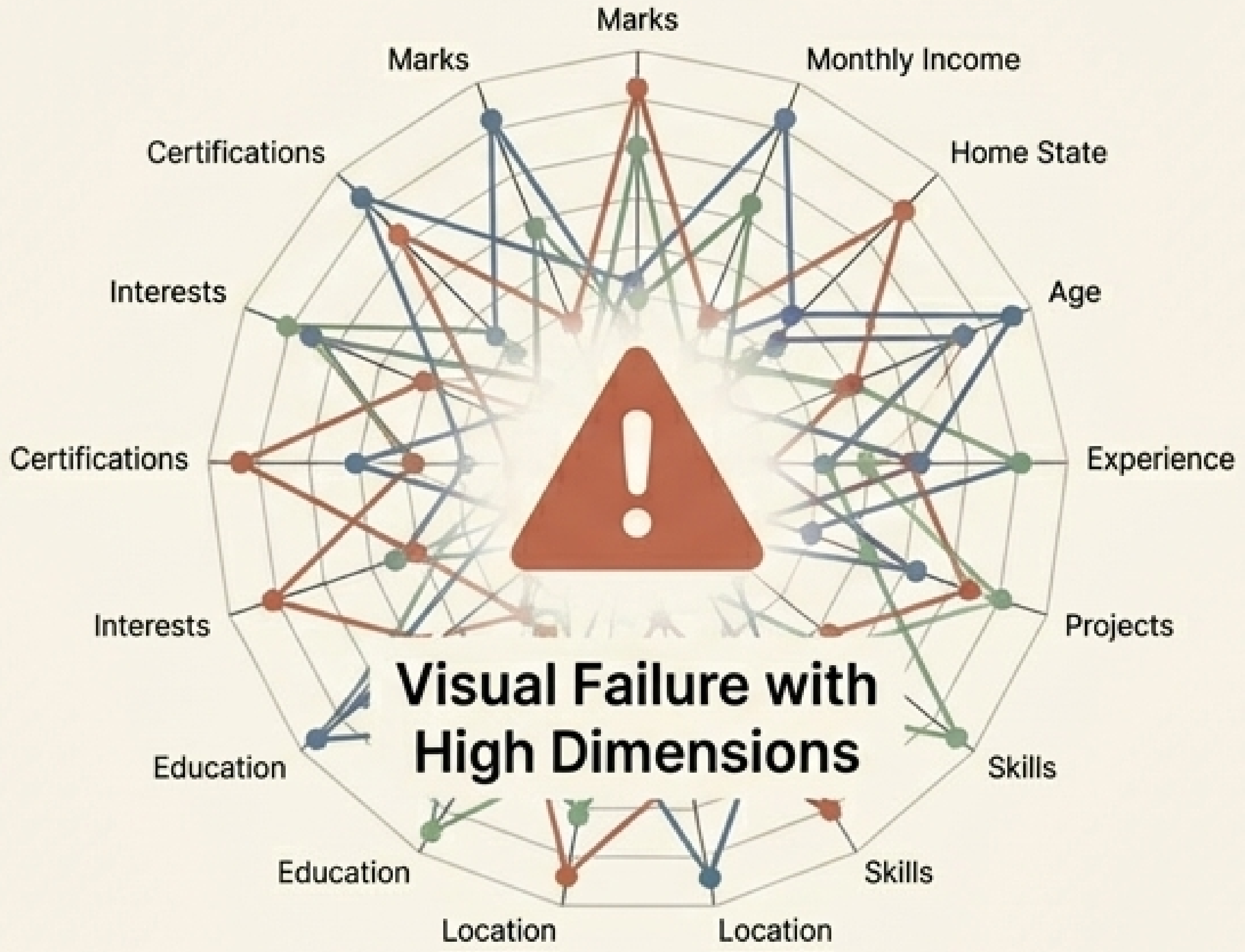
**The Goal:** Group 4,000 students effectively to tailor their placement preparation.

**The Data:** We only have two metrics per student: CGPA and IQ.



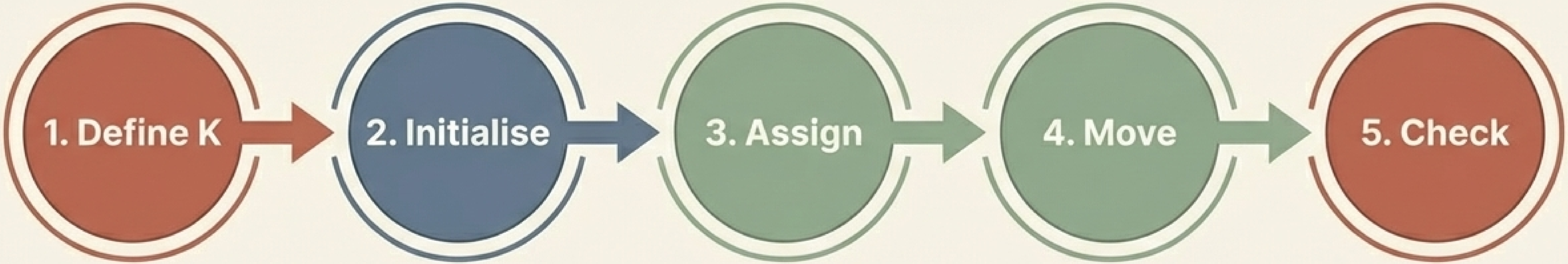
# The trap of multi-dimensional data




Human eyes can cluster 2D data. We completely fail at 10 to 15 variables. To scale, we need mathematical precision over visual estimation.

# K-Means relies on pure geometric distance




- K-Means uses Euclidean distance to find the absolute mathematical centre of data groups across n-dimensions.
- Let’s break down the internal heartbeat of the algorithm.

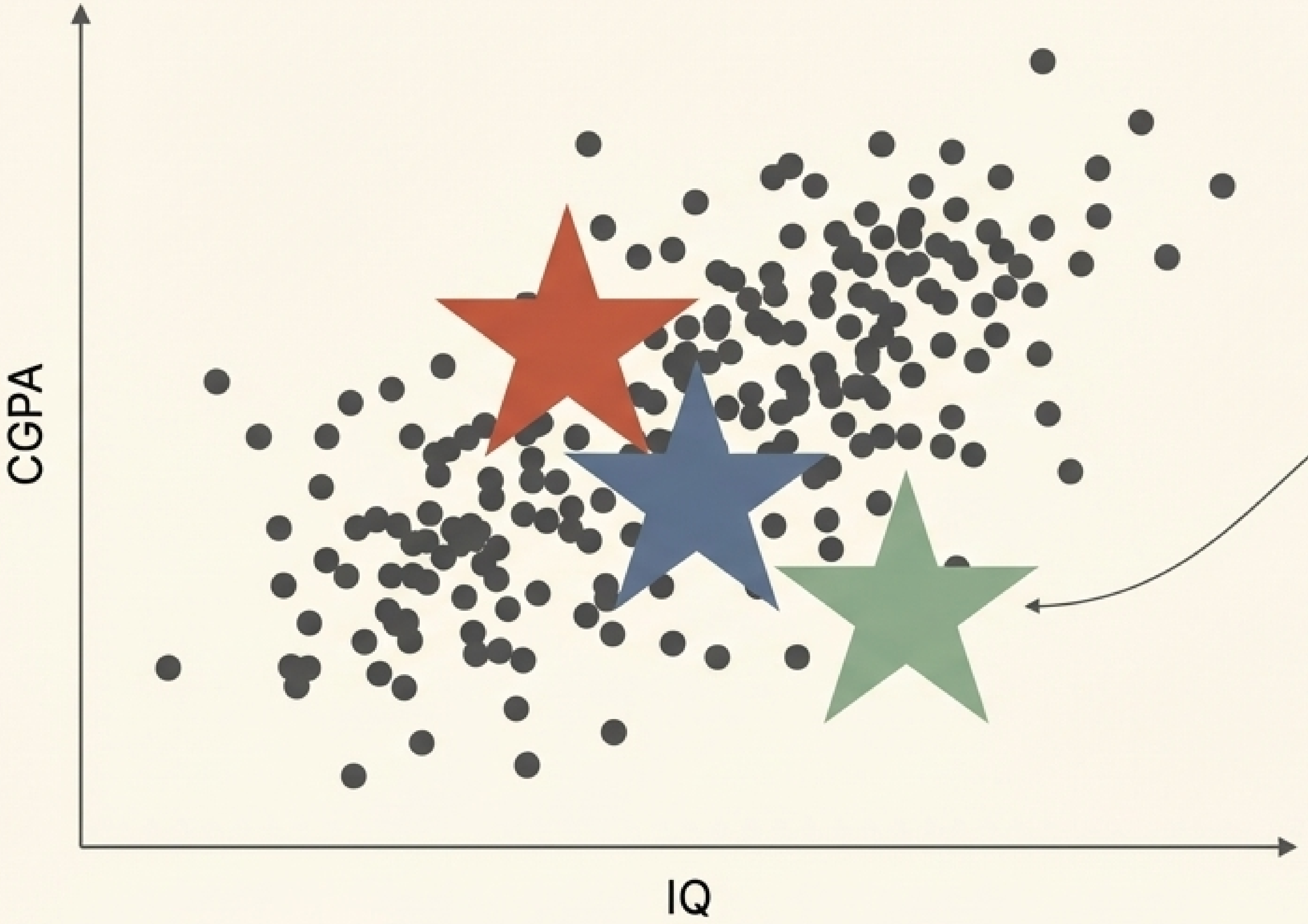
# Step 1: Define your clusters

$$K = 3$$

**The Catch**

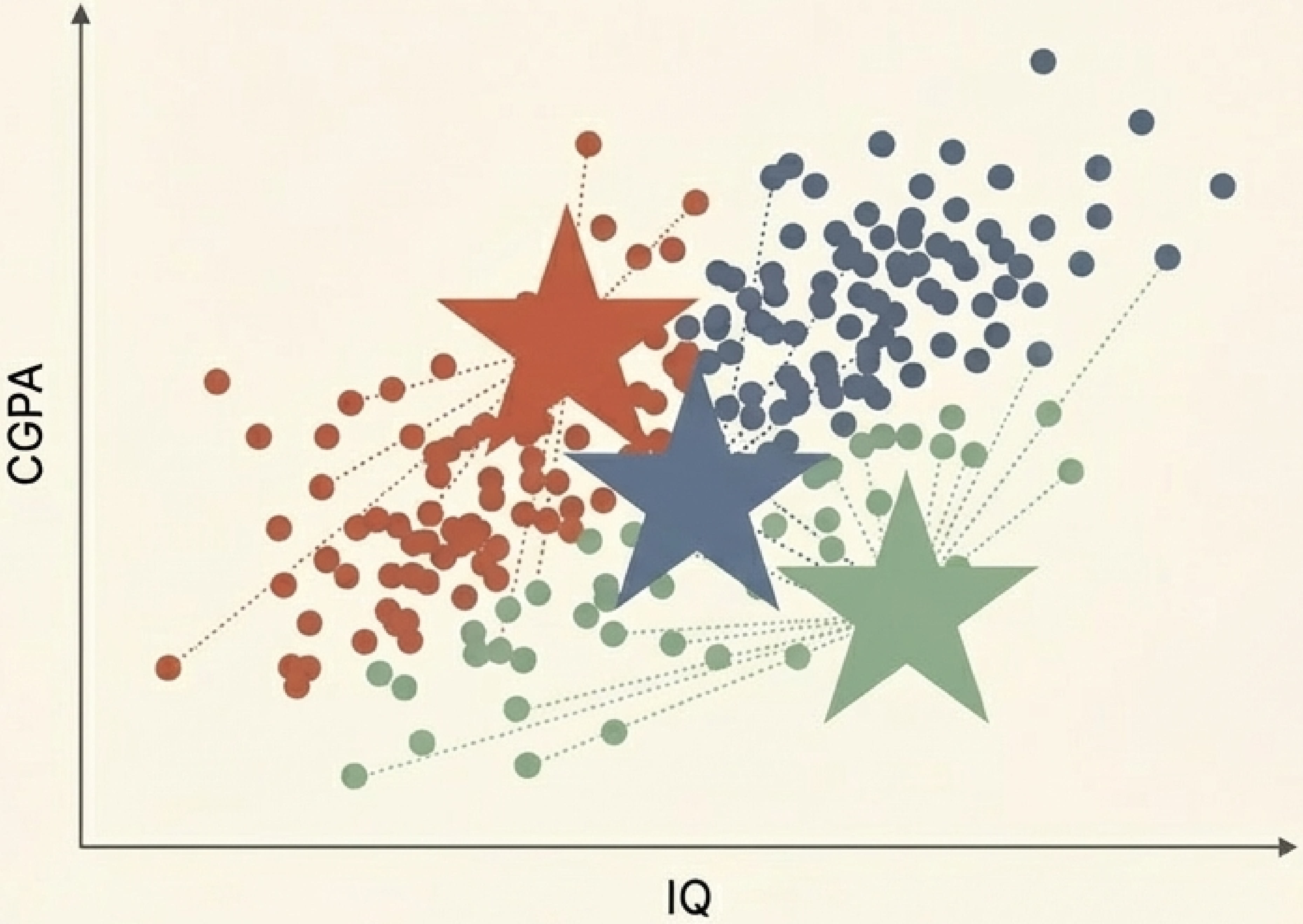
The algorithm cannot inherently know how many clusters exist. For now, we will assume K=3. We will solve the mystery of finding the perfect K at the end.

# Step 2: Randomly initialise the Centroids



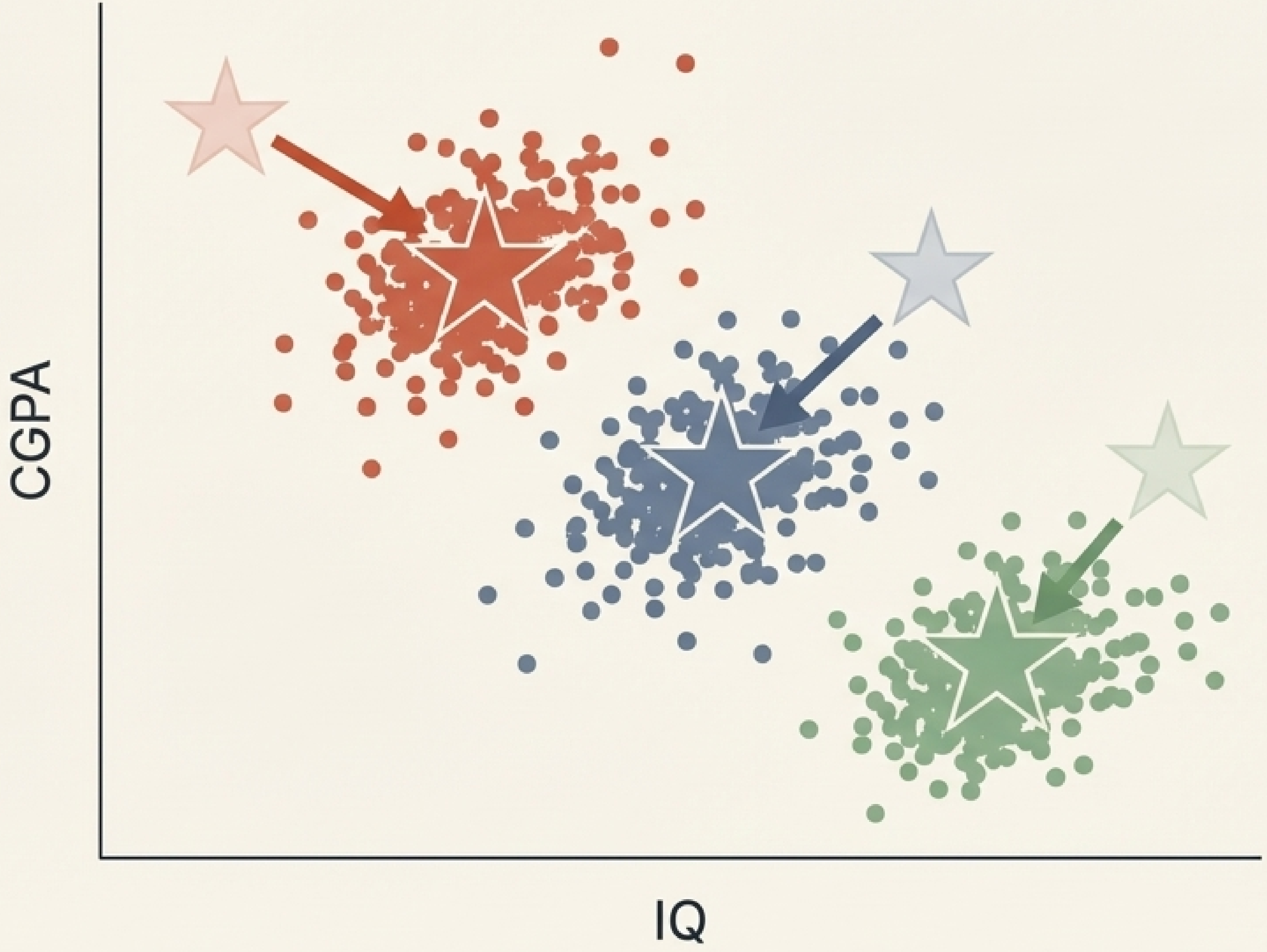

**Centroid**: The focal centre of a cluster. Initially, the algorithm picks any random coordinates to place these K centroids.

# Step 3: Assign points to the nearest centre



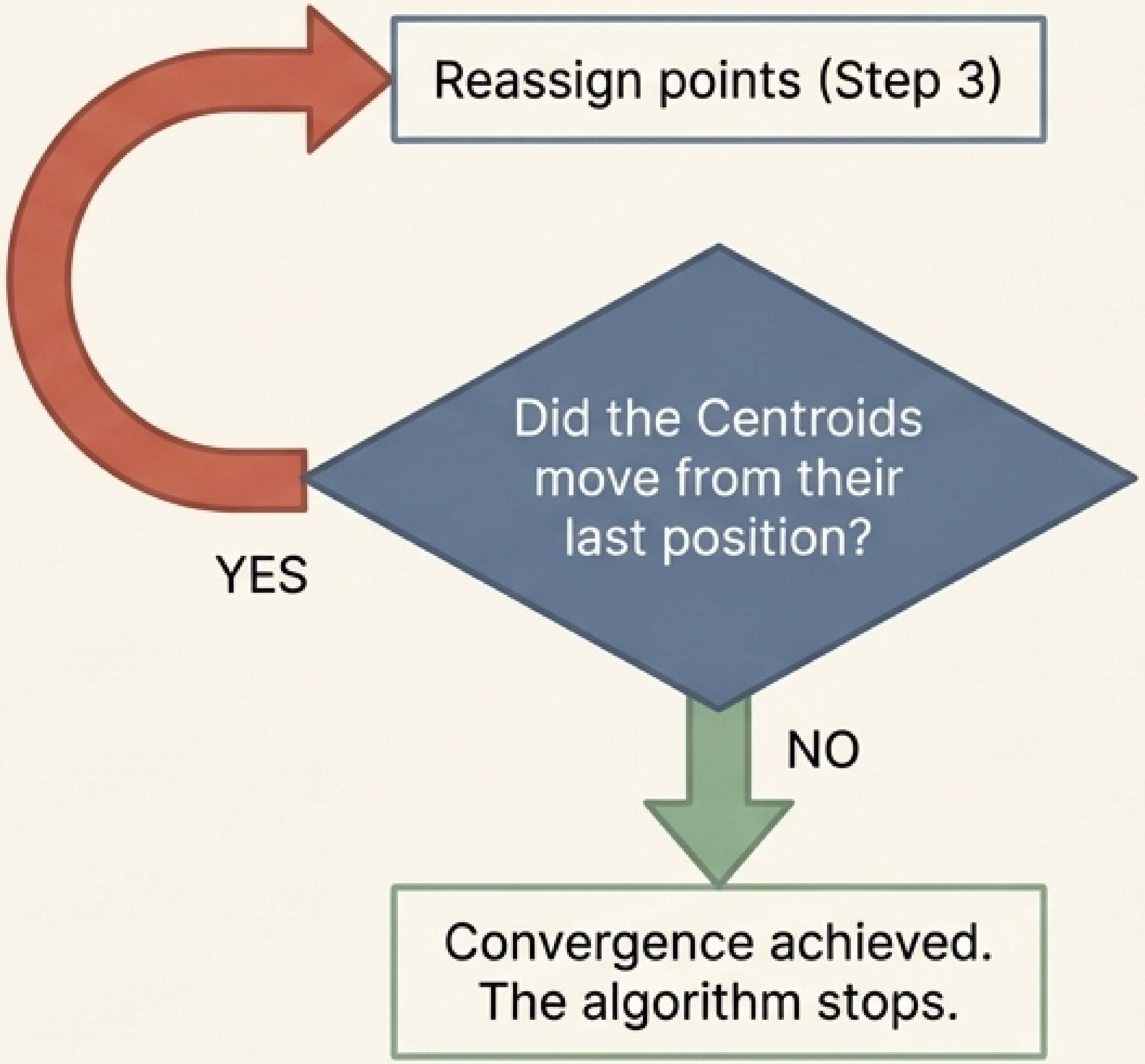

Every single student is assigned to their nearest Centroid using standard Euclidean distance. Our first, imperfect clusters are born.

# Step 4: Move the Centroids to the true average




The algorithm calculates the mean (average) CGPA and IQ for each new group.
The Centroid shifts its coordinates to this new true centre.

# Step 5: Check for convergence and repeat



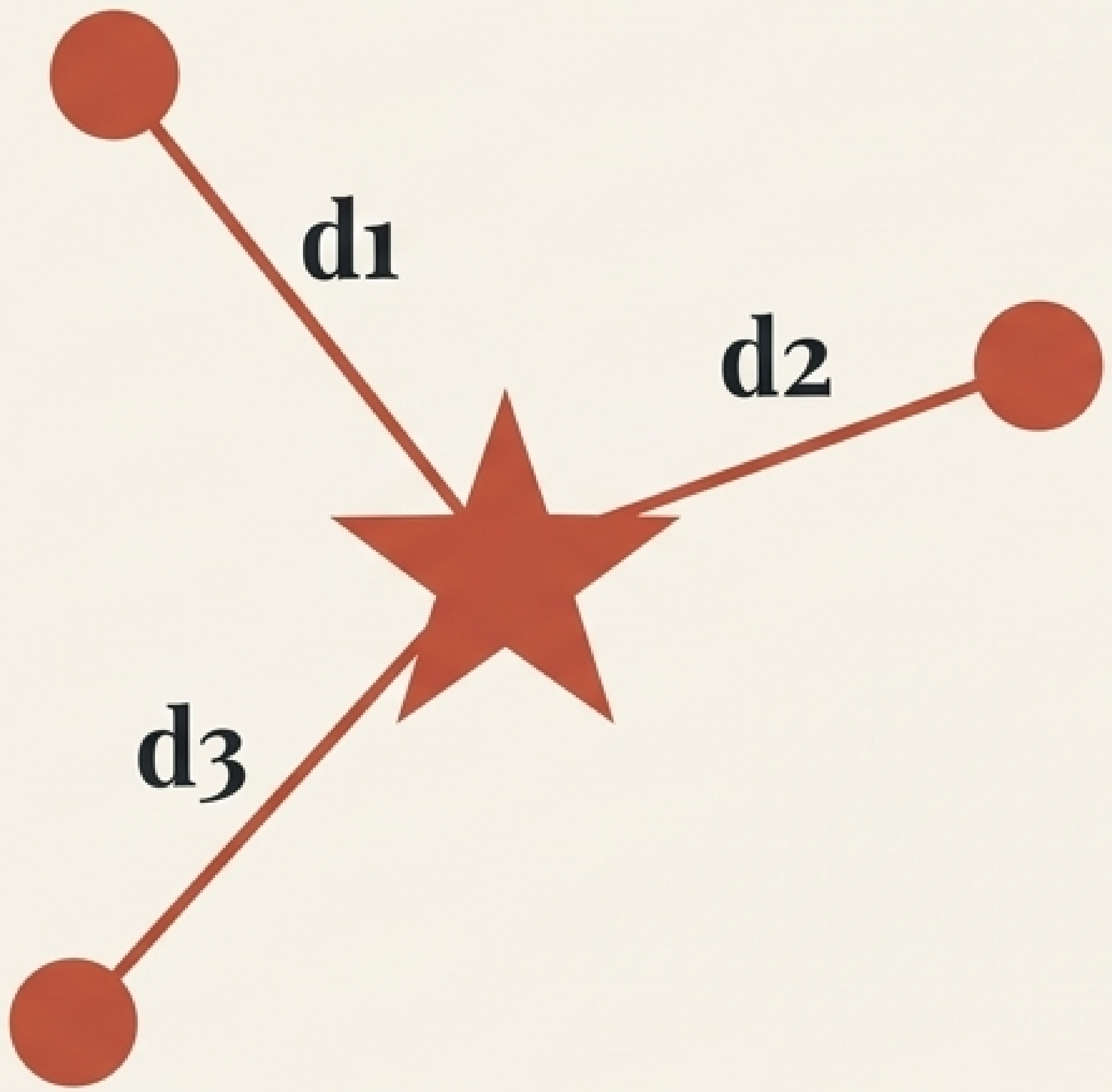

**Once the Centroids stop moving, the data is perfectly clustered.**

# The missing piece of the puzzle

$$K = 3\,?$$

We successfully clustered our 4,000 students. But how did we know to use 3 clusters?  What if there were actually 4, or 5? We need a mathematical way to find the optimal number of groups.

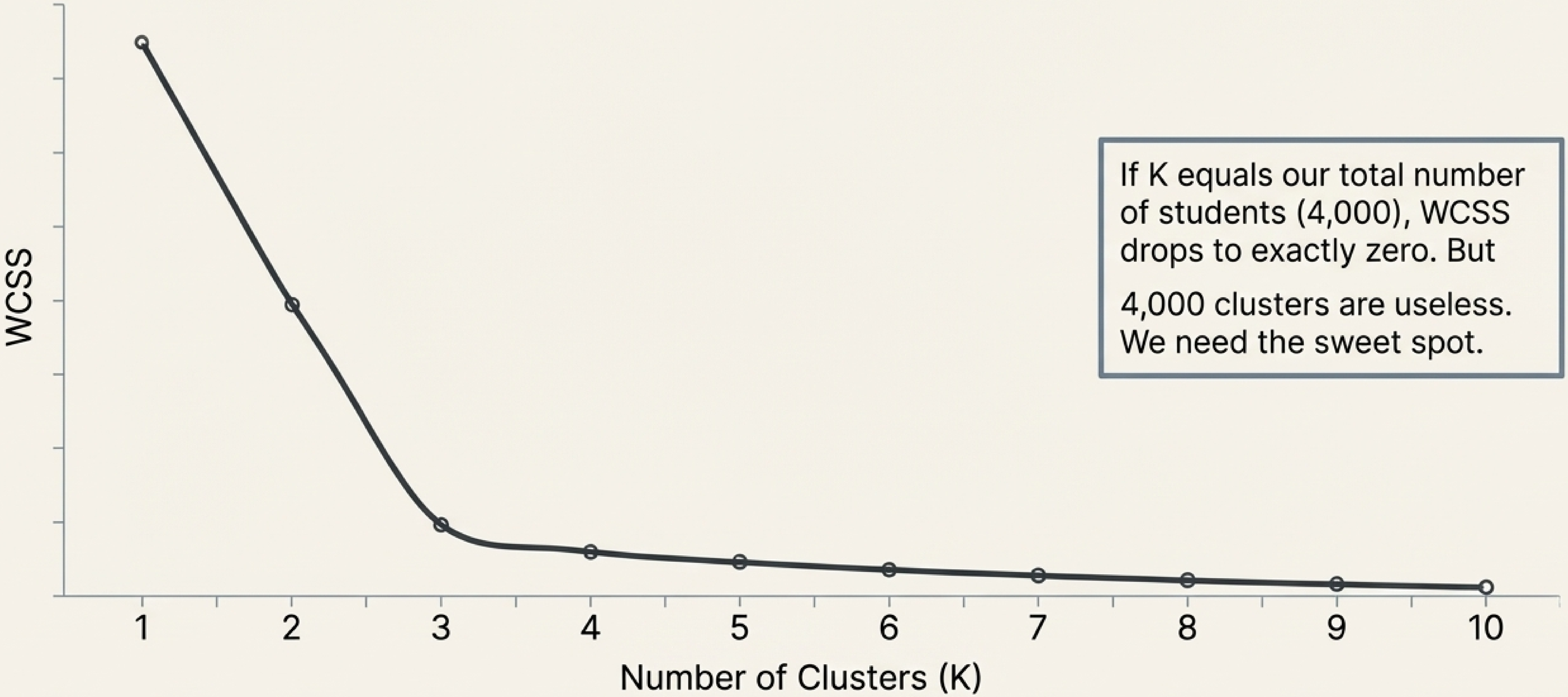
# Measuring error with WCSS




$$WCSS = d_1^2 + d_2^2 + d_3^2$$

- WCSS (Within-Cluster Sum of Squares): Also known as Inertia.
- It measures the squared distance from a centroid to all its assigned points.
- The Objective: Minimise this number. Tighter clusters mean lower WCSS.

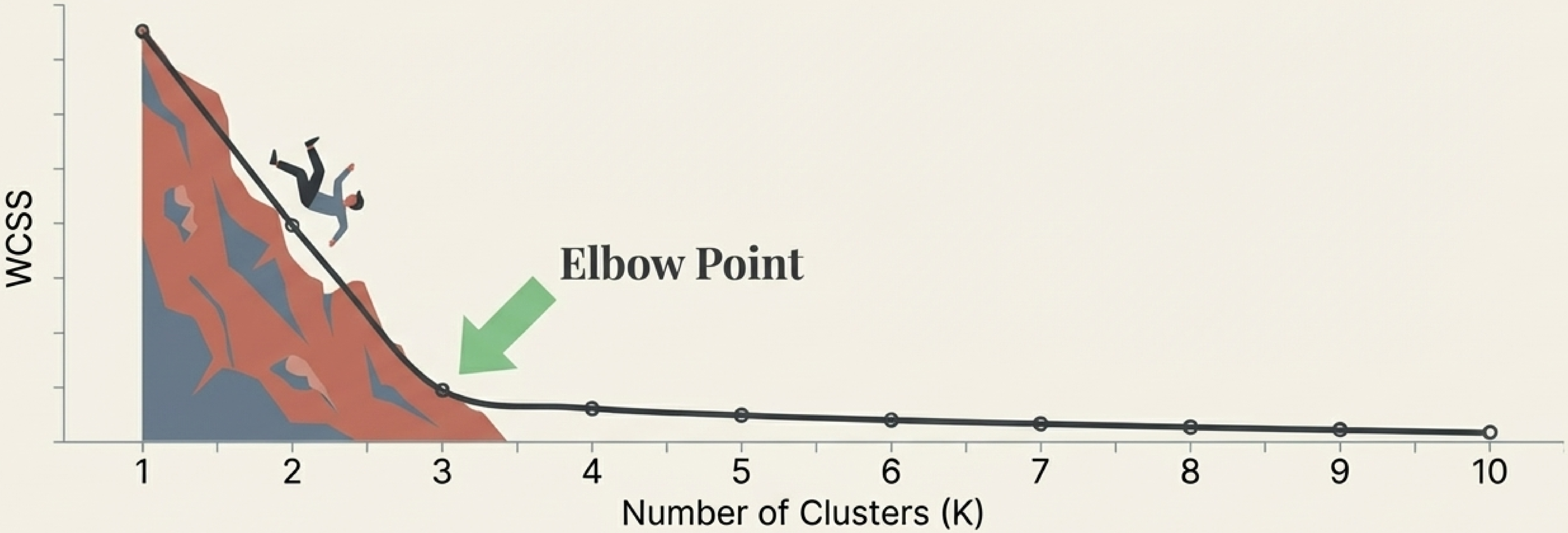
# Plotting the Elbow Curve




If K equals our total number of students (4,000), WCSS drops to exactly zero. But

4,000 clusters are useless. We need the sweet spot.

# Finding the point of diminishing returns




- Imagine falling down a mountain. At first, the drop is steep and fast (massive WCSS reduction).

- The Elbow Point (K=3) is where the terrain flattens out.

- Moving from 3 to 4 clusters yields negligible benefits. K=3 is our mathematically optimal answer.

# The elegance of geometric clustering

**Scalable:** Relies entirely on simple Euclidean distance, making it mathematically viable across infinite dimensions.

**Iterative:** Self-corrects through a continuous loop of assigning and moving centroids.

**Optimised:** The Elbow Method ensures we build structures based on mathematical efficiency, not human guesswork.